Paper 16 — Why Width Can Make Bad Minima Exceptional

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The Loss Surface of Deep and Wide Neural Networks

Key: `nguyen2017loss` | Tags: G / A / D; F / B | Tier: Must read (main theorems)

The paper in one minute

Question: Does a wide neural network still have suboptimal local minima, or does extra width make its loss geometry more favorable?
Model/Method: Fully connected networks with smooth analytic activations; analyze rank of hidden-layer features, backpropagation gradients, and non-degenerate Hessian blocks.
Finding: With a layer of at least $N-1$ units for N distinct training examples and suitable downstream rank and non-degeneracy conditions, the covered critical points—including the covered local minima—are globally optimal.
Why we care: Architecture can rule out *certain bad stationary points* without requiring a new optimizer. Width alone is not a universal no-bad-minima theorem.

1. Start with the question: why can extra width help?

A network can fit the training examples poorly even though its parameters are at a stationary point ($\nabla\Phi = 0$). The authors ask whether a sufficiently expressive hidden layer prevents this *at regular stationary points*. They focus on the **geometry of the training loss**, not on how many gradient-descent iterations are needed.

Let N be the number of training samples, n_k the width of hidden layer k, and $F_k \in \mathbb{R}^{N\times n_k}$ the matrix whose ith row is the hidden representation of example i. Append a bias column $\mathbf{1}$:

$$A = [F_k\ \mathbf{1}] \in \mathbb{R}^{N\times(n_k+1)}.$$

If $\operatorname{rank}(A) = N$, its rows are independent: there is enough linear freedom at that layer to distinguish all N training examples.

2. Worked example: three examples, only two hidden units

This is a *teaching example of the feature-rank mechanism*, with a linear readout for transparent algebra. The paper proves a more general statement for its specified multilayer networks and activations.

Take inputs $x = -1, 0, 1$ and two tanh units, $h_1(x) = \tanh(x)$ and $h_2(x) = \tanh(2x+1)$. The augmented feature matrix is

$$A \approx \begin{bmatrix} -.762 & -.762 & 1 \\ 0 & .762 & 1 \\ .762 & .995 & 1 \end{bmatrix}, \qquad \det A \approx -0.982 \neq 0.$$

For desired targets $y = (1, 0, 2)^\top$, solving $A\beta = y$ gives

$$\beta \approx (3.339, -2.326, 1.772)^\top, \qquad A\beta \approx y.$$

So two hidden features *plus a bias* can independently fit three labels. This illustrates why the width threshold is $n_k \geq N-1$, not necessarily $n_k \geq N$.

Why this example matters. In a simple squared-loss readout, $\nabla_\beta \frac{1}{2}\|A\beta - y\|^2 = A^\top(A\beta - y)$. If A has full row rank, $A^\top r = 0$ forces $r = 0$. Thus a stationary readout cannot keep nonzero training residuals. The paper extends this intuition through backpropagation and rank conditions on later layers.

3. The main theorem: exactly what is guaranteed?

Theorem 3.8 (informal but faithful). Assume the paper's analytic-activation and loss conditions. Suppose some hidden layer k has

$$n_k \geq N - 1.$$

A *critical point* is a global minimizer if it additionally satisfies:

1. A Hessian **block** involving parameters from an upper-layer subset that *includes layer* $k+1$ is non-singular (the paper's block non-degeneracy condition).
2. Every weight matrix from layer $k+2$ through the output has **full column rank**: $\operatorname{rank}(W_\ell) = n_\ell$.

Because a matrix can have full column rank only if it has at least as many rows as columns, condition 2 implicitly requires a **pyramidal tail**: $n_{k+1} \geq n_{k+2} \geq \cdots \geq n_L$. *Corollary 3.9* gives the corresponding statement for non-degenerate local minima with full-rank upper layers.

> **Be precise about "almost all."** The authors prove that the set of *hidden-layer parameters producing deficient augmented feature rank* has Lebesgue measure zero (Lemma 4.4). Their main global-optimality conclusion still requires Hessian-block non-degeneracy and upper-layer full rank. They explicitly do **not** exclude all degenerate or rank-deficient spurious local minima. Do not paraphrase this as "every local minimum is global."

4. Understand the conditions rather than memorize them

Condition	Intuition / why the proof uses it
N distinct examples; analytic, strictly monotone activation	Makes full-rank feature representations possible except for exceptional parameter choices; sigmoid, tanh, softplus are covered.
$n_k \geq N - 1$	The n_k features plus the bias can span N independent training-example directions.
Full column rank after $k+1$	The downstream network does not destroy error information through a rank-deficient map.
Non-singular upper-layer Hessian block	Lets the implicit function theorem track stationary upper-layer parameters under small changes to earlier layers.
Appropriate smooth loss; minimum attained	Critical loss residuals must correspond to global minima of the per-example loss.

Not included in the core guarantee: nonsmooth ReLU itself, convolutions with sparse connectivity, cross-entropy without checking the assumptions, and any generic GD/SGD convergence rate.

5. Why use the Hessian if this is not a saddle-escape paper?

Here the Hessian is *not* used to design Newton updates. Its invertible upper-layer block is a proof tool: it permits upper-layer stationary solutions to vary smoothly when earlier layers change. This is different from the negative-eigenvalue arguments in saddle-escape papers.

6. Proof intuition: four logical moves

The proof is advanced, but you should retain its structure rather than its technical details.

1. **Rank implies zero training error at a suitable critical point.** Lemma 3.5 says that if $[F_k, \mathbf{1}]$ has rank N, the later matrices have full column rank, and the gradient at layer $k+1$ vanishes, then the point is globally optimal.
2. **Width makes full feature rank generic.** Analyticity implies that a nontrivial determinant-type expression does not vanish on a positive-volume parameter region. Lemmas 4.3–4.4 establish that rank-deficient feature parameters form a measure-zero set.
3. **Move slightly while remaining stationary above.** The non-singular upper Hessian block permits an application of the implicit function theorem: small lower-layer changes can be paired with adjusted upper-layer parameters that retain upper-layer stationarity.
4. **Transfer global optimality back to the original point.** Nearby full-feature-rank stationary configurations are globally optimal; continuity and local stationary structure show that the original covered critical point is global too.

Think of a locked door. A poorly represented set of examples might seem to block progress; width says the representation can be nudged to distinguish all examples. Full downstream rank carries that information forward, and Hessian non-degeneracy keeps the stationarity equations well behaved under that nudge. The original point cannot then be a regular bad local minimum.

7. A second result you should not confuse with the first

Theorem 5.3: linear separability of learned features. The authors also study a *different class-dependent loss*, akin to squared hinge loss. If examples have become linearly separable at a layer and later matrices have full column rank, then a critical point is globally optimal for that classification objective. This does *not* automatically mean that linear separability alone gives zero squared regression loss, nor is it the same statement as Theorem 3.8.

What does “pyramidal” mean?

Suppose a network has layer widths $[3, 80, 25, 8, 2]$. From the 80-unit layer onward, widths decrease. A chain like $[3, 80, 10, 40, 2]$ is *not* pyramidal after the wide layer; rank preservation in the expanding $10 \to 40$ step is impossible if its weight matrix is required to have full column rank.

What the authors leave open

The theorem does not show that *standard gradient descent* or SGD actually reaches the regular critical points, how quickly it does so, or whether its generalization is good. It also leaves possible degenerate and low-rank suboptimal minima unruled out. The discussion explicitly raises sparse-connectivity extensions as future work.

8. Main contributions, assumptions, and limitations

Contribution	What we can honestly say in the survey
Benign wide-network geometry	A broad class of stationary points in sufficiently wide, smooth, fully connected networks is globally optimal, given additional rank and non-degeneracy conditions.
Generic feature rank	With $n_k \geq N-1$, deficient augmented hidden-feature rank occurs only on a measure-zero set of lower-layer parameters.
Multilayer extension	Hidden-layer feature rank and full-rank downstream propagation extend simpler single-hidden-layer intuitions.
Feature separability result	A separate classification result covers critical points at which learned hidden features are linearly separable.

Important limits. No universal "all minima are global" theorem; no proof that degenerate/rank-deficient bad minima cannot exist; smooth analytic activations rather than ReLU itself; an often demanding width threshold $n_k \geq N-1$; the result is about loss geometry, not training time, SGD guarantees, or generalization.

9. Connection to papers we already studied

Paper	The distinction to remember
Safran & Shamir (2018)	Finds certified spurious minima in selected finite-width teacher–student ReLU population losses; Nguyen–Hein proves good geometry for a different, wide, smooth empirical-loss setting. No contradiction.
Ge (2017); Bhojanapalli (2016)	Obtain benign landscapes for *structured matrix recovery*; Nguyen–Hein analyzes deep neural-network architectures directly.
Du (2019); Allen-Zhu (2019)	Prove optimization convergence for specified overparameterized networks. Nguyen–Hein mainly classifies critical points; a benign landscape does not itself supply a rate or successful algorithm.
Lee (2016); Jin (2017)	Study avoidance or finite-time escape from saddles; Nguyen–Hein characterizes which regular stationary points can be globally optimal.

10. Quick reference and reading path

Term	Meaning / what to remember
N; n_k	Training examples; width of hidden layer k. Main width condition: $n_k \geq N-1$.
F_k; $[F_k, \mathbf{1}]$	Hidden-feature matrix; augmented matrix must have *row rank N* for the simple gradient argument.
Critical point	All parameter gradients vanish; may be a minimum, saddle, or maximum.
Non-degenerate Hessian block	An invertible Hessian submatrix for selected upper-layer parameters.
Theorem 3.8 / Corollary 3.9	Under the full set of conditions, critical points / non-degenerate local minima are global.
Theorem 5.3	Separate result for classification loss and linear separability of learned features.

Must read: Abstract, Section 3.2 (Assumptions 3.2, Lemma 3.5, Theorem 3.8, Corollary 3.9), Section 4 (Lemmas 4.3–4.4 and proof idea), Sections 5–6. **Skim:** long implicit-function and matrix-rank proof details.

Papers Master — Our notes (copy): Nguyen and Hein (2017) show that in fully connected networks with analytic monotone activations, a layer with at least $N-1$ units makes augmented hidden features generically full rank. Under upper-layer full-rank and Hessian-block non-degeneracy assumptions, the covered critical points are global minima. Does not rule out degenerate/rank-deficient bad minima or prove GD/SGD convergence rates.

Source: Nguyen & Hein, ICML 2017, original PDF. The 3-example interpolation is a pedagogical adaptation, not a reported experiment.